

INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN: AN AGILE, ITERATIVE APPROACH

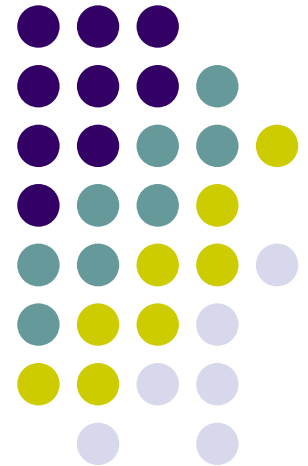
SATZINGER | JACKSON | BURD

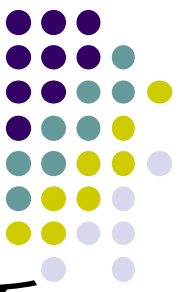
CHAPTER 7

Designing the User and System Interfaces

Chapter 7

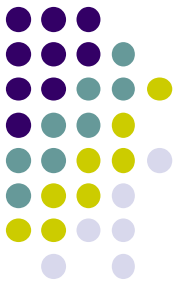
Introduction to Systems
Analysis and Design:
An Agile, Iterative Approach
6th Ed
Satzinger, Jackson & Burd





Chapter 7 Outline

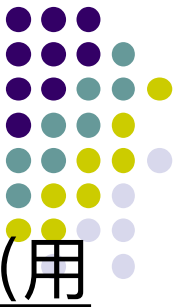
- User and System Interfaces (用户界面和系统界面)
- Understanding the User Interface (理解用户界面)
- User-Interface Design Concepts (用户界面的设计概念)
- The Transition from Analysis to User-Interface Design (从分析到用户界面设计的转换)



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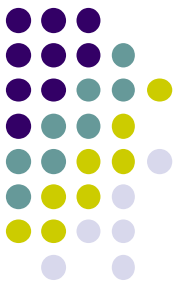
- User-Interface Design (用户界面设计)
- Identifying System Interfaces (确定系统界面)
- Designing System Inputs (设计系统输入)
- Designing System Outputs (设计系统输出)

Learning Objectives



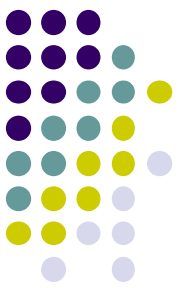
- Describe the difference between user interfaces (用户界面) and system interfaces (系统界面)
- Describe the historical development of the field of human-computer interaction (人机交互领域的发展)
- Discuss how visibility (可视性) and affordance (提示性) affect usability
- Describe user-interface guidelines (用户界面的指导规则) that apply to all types of user-interface types and additional guidelines specific to Web pages and mobile applications

Learning Objectives (continued)

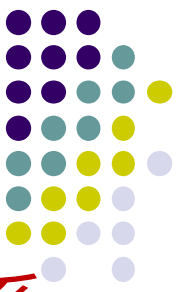


- Create storyboards (创建故事脚本) to show the sequence of forms used in a dialog
- Discuss examples of system interfaces found in information systems (讨论系统接口)
- Define system inputs and outputs (定义输入输出) based on the requirements of the application program
- Design printed and on-screen reports appropriate for recipients (设计适合收件人的报告)

Overview



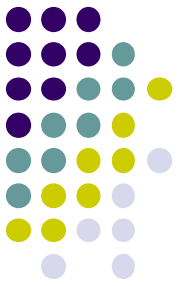
- Information systems interact with people and other systems (信息系统与人和其他系统相互作用)
- Poorly designed user interface can make the information system unusable (设计不良用户界面会使信息系统无法使用)
- Poorly designed system interfaces are a source of errors and inefficiency (设计不良的系统接口是错误和低效率的根源)
- User and system interfaces involve both inputs and outputs and involve a large number of stakeholders (用户和系统接口涉及输入和输出，并涉及大量利益相关者)



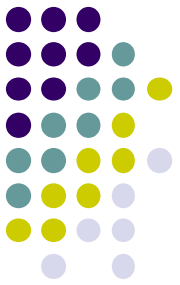
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User Interfaces and System Interfaces

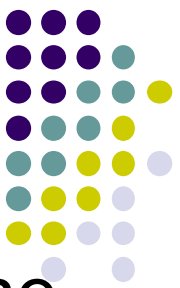


- User Interface (用户界面) – inputs and outputs that directly **involve a human user/actor** (直接涉及人类用户的输入输出)
- System Interface (系统接口) – the inputs and outputs that **require minimal human intervention** (输入和输出需要最少的人为干预)
 - Inputs captured automatically (自动捕获输入)
 - Outputs direct to other systems (直接输出到其他系统)
 - Printed and distributed outputs (打印和分发输出)



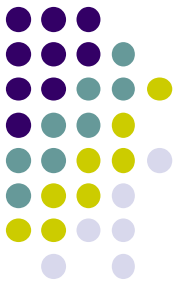
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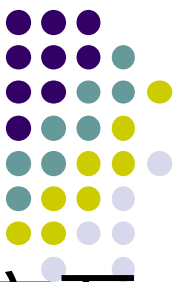
User Centered Design

- Design techniques that embody the view that the user interface is the system to the user (从用户的角度看，用户界面就是整个系统。界面后的程序、数据库、硬件都是不相关的)
- Principles of User Centered Design (三个重要原则)
 - Focus early on users and their work (及早关注用户及其工作)
 - Evaluate designs to ensure usability (评价设计以确保其可用性)
 - Use iterative development (使用迭代开发方法)



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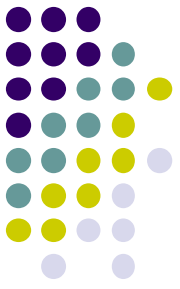


User Interface Design Concepts

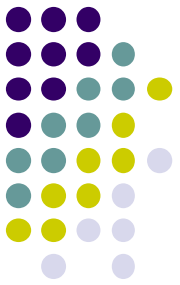
- Human-Computer Interaction (HCI) 人机交互
 - A field of study concerned with the efficiency and effectiveness of user interaction with computer systems, human-oriented input and output technology, and psychological aspects of user interfaces
- Visibility and Affordance 可视性和提示性
 - To be usable, a control must be visible to the user and its appearance should suggest its functionality



User Interface Design Guidelines



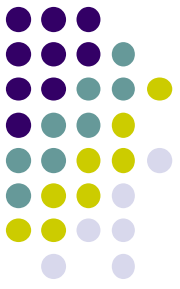
- Design for Consistency (一致性)
- Provide Shortcuts (快捷方式)
- Provide Feedback (反馈)
- Dialogs Should Yield Closure (完整的对话)
- Error Handling that Provides Guidance (错误处理)
- Easy Reversal of Actions (撤销)
- Reduce Short Term Memory Load (减轻短期记忆负担)



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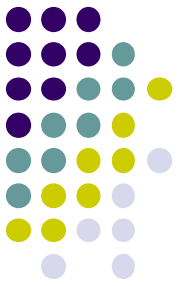
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Use Cases and the Menu Hierarchy



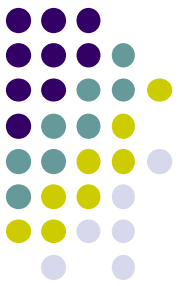
- Menus are a typical way to organize access to use case functionality (菜单是组织对用例功能访问的典型方式)
- Different types of users might have different menus (不同类型的用户可能有不同的菜单)
- Useful to design an overall menu hierarchy and then subsets for different users (可用于设计整体菜单层次结构，然后为不同用户设计子集)

Guidelines for Windows and Forms



- Interface Layout and Formatting (设计窗体和表格)
 - Consistency, labels and headings, distribution and order, fonts and colors (一致性、标签和标题、分布和顺序、字体与颜色)
- Data Entry (数据输入)
 - Text box, list box, combo box, radio buttons, check boxes (文本框、列表框、组合框、单选按钮、复选框)
- Navigation and Support Controls (导航)
 - Minimize, maximize, close, scroll bars, resize (最大化、最小化、关闭、滚动条、调整大小)

RMO Windows Form



RMO Customer Support System - Product Detail



**RIDGELINE MOUNTAIN
OUTFITTERS**

Product Information

Product ID

Size

Color

Product Picture



Next/Previous Picture

Product Description

Toddlers medium-weight parka. Fleece lined. Hood with velcro closure. Elastic sleeve openings with glove/mitten hooks. One interior and two exterior pockets with velcro closures. Machine wash and dry. Nylon/polyester shell. Cotton lining.

Regular Price

Sale Price

Inventory

ID

In Stock

On Order

Due Date

Search Criteria

Key Words

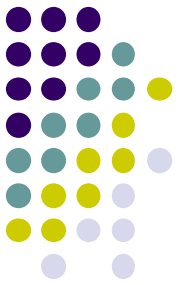
Catalog

Gender

Product Type

Price
Min
Max

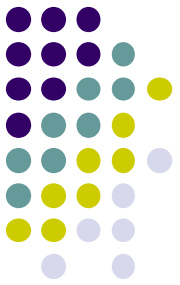
Matches



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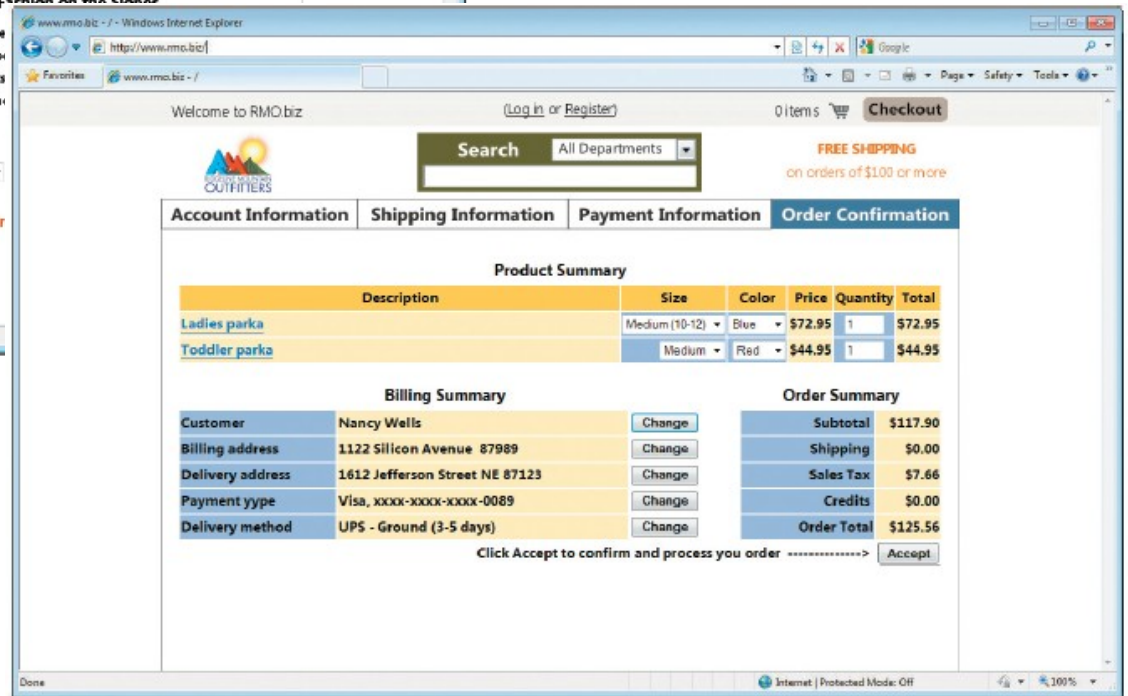
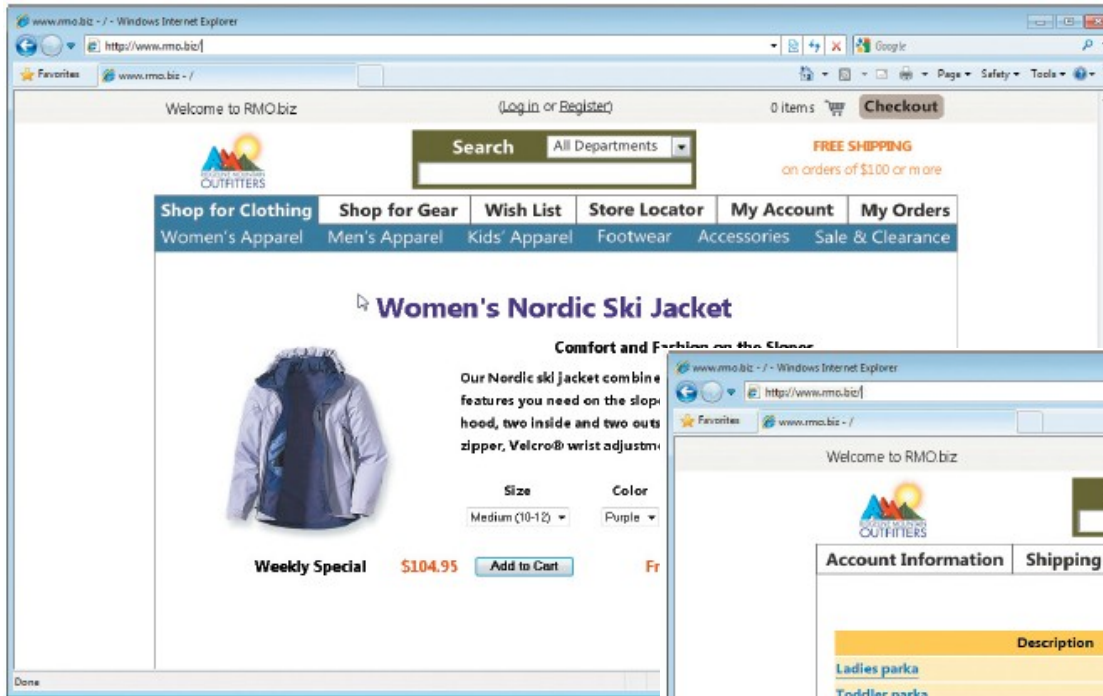
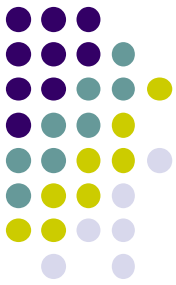
Guidelines for Web Browser User Interfaces



- Consistency (一致性)
 - Cascading Style Sheets (CSS) – Web page encoding standard that enables a Web site designer to specify parts of a page that will always look the same and parts that will vary by task or audience
- Performance Considerations (性能考虑)
 - Sensitive to network connection, amount of information transmitted, type of information transmitted
- Pictures, Video, and Sound (图片、视频和声音)
 - Powerful, but compatibility issues arise
- Users with Disabilities (残疾用户)
 - Assistive technologies -- software (such as text-to-speech and voice-recognition utilities) that adapts user interfaces to the special needs of persons with disabilities

RMO Example

Using CSS for Consistency

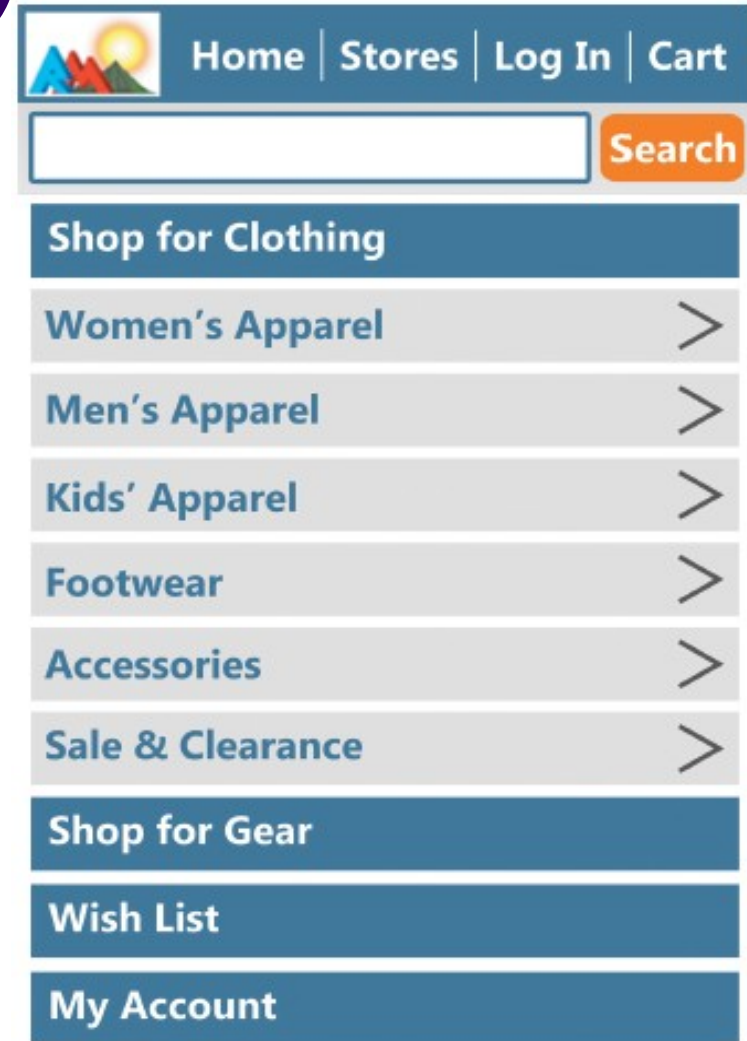


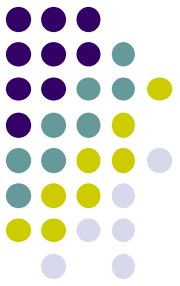
Guidelines for Handheld Devices (手持设备附加原则)



- Challenges

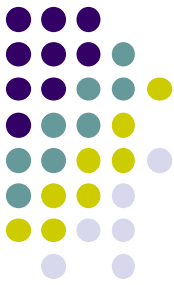
- Small screen size, small keyboards and touch screens, limited network capacity, app design guidelines and toolkits (小屏幕、小键盘、触摸屏、有限的网络容量)





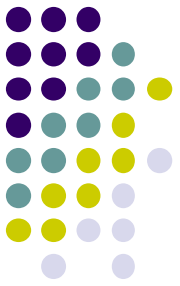
人机界面设计指南

- ① 一般交互指南：涉及信息显示、数据输入和系统整体控制
 - (1) 保持一致性。
 - (2) 提供有意义的反馈。
 - (3) 在执行有较大破坏性的动作之前要求用户确认。
 - (4) 允许取消绝大多数操作。
 - (5) 减少在两次操作之间必须记忆的信息量。
 - (6) 提高对话、移动和思考的效率。
- ① 允许犯错误。
- ② 按功能对动作分类，并据此设计屏幕布局。
 - (1) 提供对用户工作内容敏感的帮助设施
 - (2) 用简单动词或动词短语作为命令名。



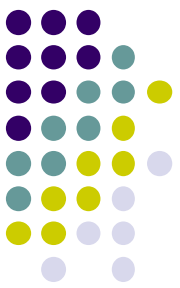
人机界面设计指南

- ② **信息显示指南：**多种不同方式“显示”信息：用文字、图形和声音；按位置、移动和大小；使用颜色、分辨率和省略。
 - (1) 只显示与当前工作内容有关的信息。
 - (2) 不要用数据淹没用户，应该用便于用户迅速吸取信息的方式来表示数据。
 - (3) 使用一致的标记、标准的缩写和可预知的颜色。
 - (4) 允许用户保持可视化的语境。
 - (5) 产生有意义的出错信息。
 - (6) 使用大小写、缩进和文本分组以帮助理解。
- ② 使用窗口分隔不同类型的信息。
- ③ 使用“模拟”显示方式表示信息，使信息更容易被用户提取。
 - (1) 高效率地使用显示屏。



人机界面设计指南

- ③ **数据输入指南：**用户的大部分时间用在选择命令、输入数据和向系统提供输入。
 - (1) 尽量减少用户的输入动作。
 - (2) 保持信息显示和数据输入之间的一致性。
 - (3) 允许用户自定义输入。
 - (4) 交互应该是灵活的，并且可调整成用户最喜欢的输入方式。
 - (5) 使在当前动作语境中不适用的命令不起作用。
 - (6) 让用户控制交互流。

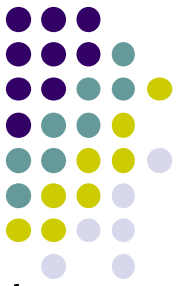


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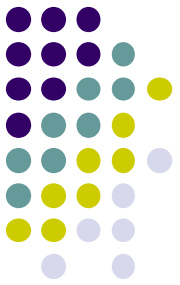
Identifying System Interfaces

Inputs/outputs with minimal human intervention

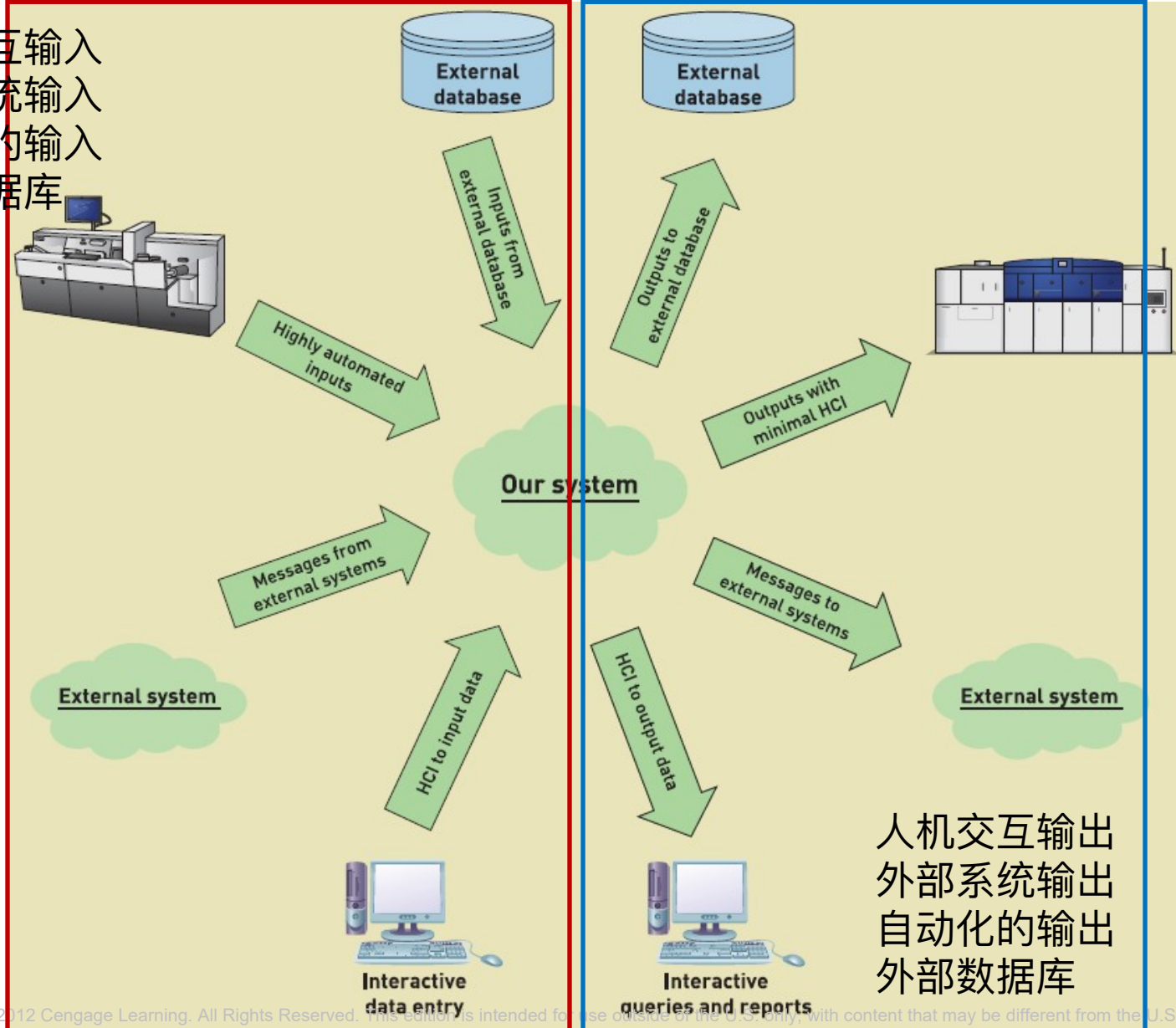


- Inputs from and outputs to other systems （与其他系统交互的输入和输出）
 - These are direct interfaces with other information systems, normally formatted as network messages.
- Highly automated inputs and outputs （高自动化的输入和输出）
 - These are captured by devices (such as scanners) or generated by persons who start a process that proceeds without further human intervention.
- Inputs and outputs to external databases （外部数据库的输入和输出）
 - These can supply input to or accept output from a system.

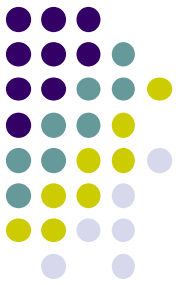
Identifying System Interfaces



人机交互输入
外部系统输入
自动化的输入
外部数据库



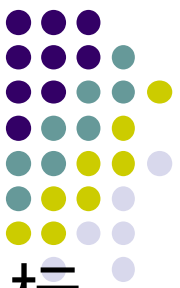
人机交互输出
外部系统输出
自动化的输出
外部数据库



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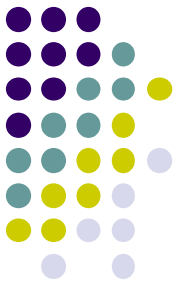
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System Inputs



- Primary Objective is Error Free Input (主要目标是无错误输入)
 - Use electronic devices wherever possible (尽可能使用电子设备和自动输入)
 - Avoid human involvement as much as possible (尽可能避免人工干预)
 - If information is already available in electronic form, use it instead of re-entering information (如果可以从某个电子表导出, 尽可能导出而不是重新输入)
 - Validate and correct information at time and location entered (在输入信息里, 对数据进行验证)

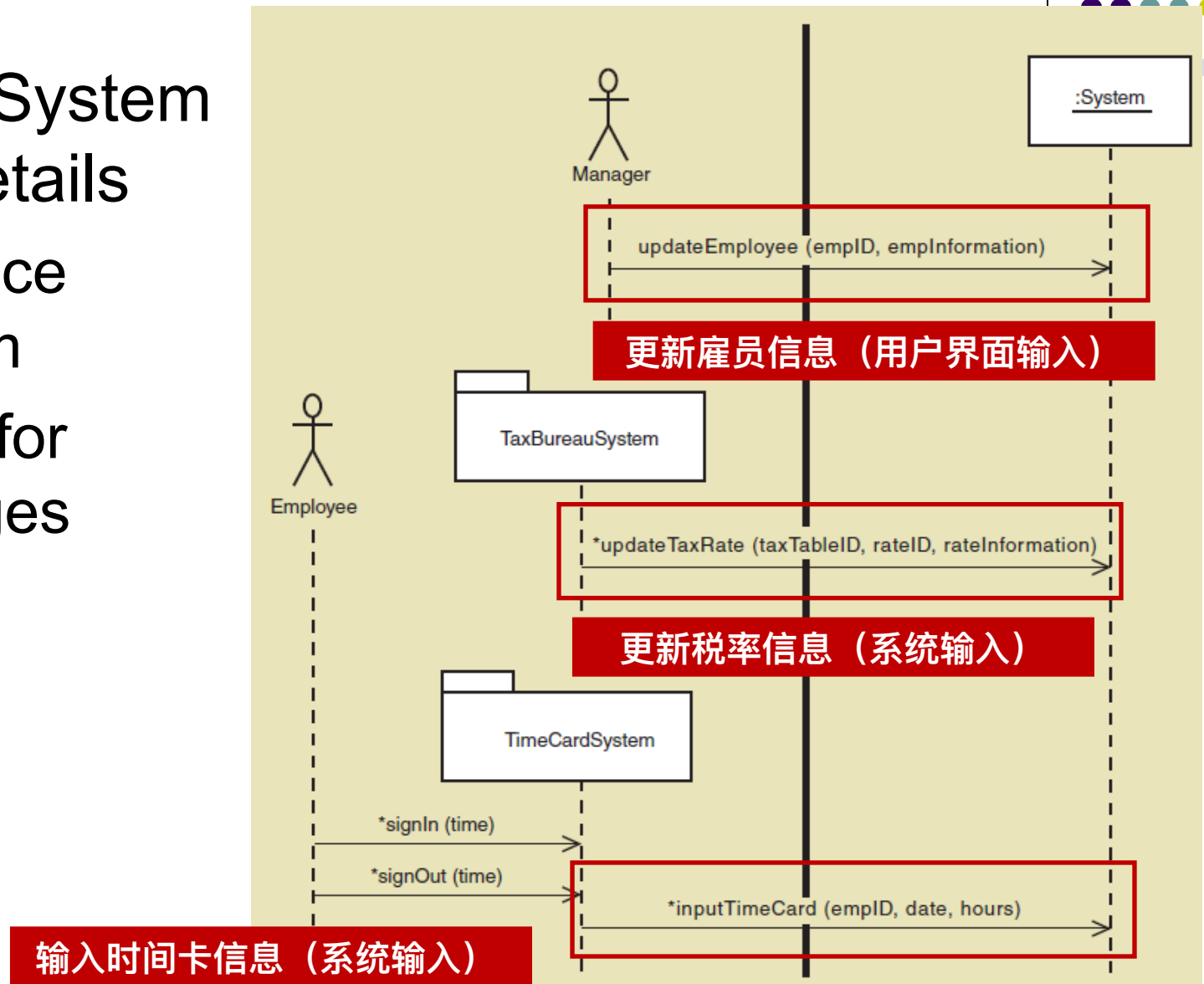
System Inputs



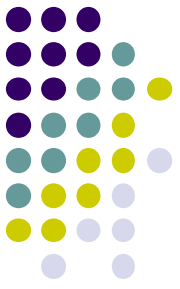
- Device Examples (设备举例)
 - Magnetic card strip readers, bar code readers, optical character recognition, radio frequency ID tags (RFID), touch screen, electronic pens, digitizers, speech recognition (磁条阅读器、条形码、光电字符识别、射频识别、触摸屏、电子笔、书写板、语音识别)



- Defining System Inputs Details
 - Sequence Diagram
 - Details for messages

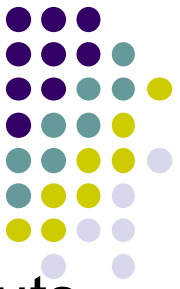


System Outputs



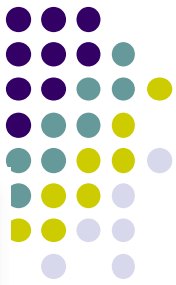
- Detailed reports (详细报告) -- reports that contain specific information on business transactions
- Summary reports (摘要报告) -- reports that summarize detail or recap periodic activity
- Exception reports (异常报告) -- reports that provide details or summary information about transactions or operating results that fall outside a predefined normal range of values
- Executive reports (执行报告) -- reports used by high level managers to assess overall organizational health and performance

System Outputs



- Internal outputs (内部输出) -- reports or other outputs produced for use within the organization
 - Types of internal reports apply here (为组织或单位内部使用而生成的, 如报表)
- External outputs (外部输出) -- reports or other outputs produced for use by people outside the organization (为组织外部人员使用而生成的, 如声明, 通知, 其他文档)
- Turnaround documents (返回文档) -- external outputs that includes one or more parts intended to be returned with new data or information (提供给用户的文档中包含了可能返回的一部分, 如账单)

External Output Example



Ridgeline Mountain Outfitters—Shopping Cart Order

Customer Name: Fred Westing
Customer Number: 6747222

Order Number: 4673064
Today's Date: May 18, 2013

Shipping Address:

936 N Swivel Street
 Hillville, Ohio 59222

Billing Address:

936 N Swivel Street
 Hillville, Ohio 59222

Qty	Product ID	Description	Size	Color	Price	Extended Price
1	458238WL	Jordan Men's Jumpman Team J	12	White/ Light Blue	\$119.99	\$119.99
1	347827OP	Woolrich Men's Backpacker Shirt	XL	Oatmeal Plaid	\$41.99	\$41.99
2	8759425SH	Nike D.R.I. – Fit Shirt	M	Black	\$30.00	\$60.00
1	5858642OR	Puma Hiking Shorts	L	Tan	\$15.00	\$15.00

Subtotal	\$236.98
Shipping	\$8.50
Tax	\$11.25
Total	\$256.73

Shipping Information:

Shipping Method: Normal 7–10 day
Shipping Company: UPS
Tracking Number: To be sent via email
Email Address: FredW253@aol.com

Payment Information:

American Express ☐ MasterCard ☐ VISA ☒ Discover ☐

Account Number

X X X X – X X X X – X X X X – 5 7 8 4

MO YR

Expiration Date 05 / 15

Thank you for your order. It is a pleasure to serve you.
 Check back next week for new weekly specials!!

Internal Output Example

Detailed control break report (控制终端详细报告)

Ridgeline Mountain Outfitters — Products and Items						
ID	Season	Category	Supplier	Unit Price	Special Price	Discontinued
RMO12587	Spr/Fall	Mens C	8201	\$39.00	\$34.95	No
Description Outdoor Nylon Jacket with Lining						
Size	Color	Style	Units in Stock	Reorder Level	Units on Order	
Small	Blue		691	150		
	Green		723	150		
	Red		569	150		
	Yellow		827	150		
Medium	Blue		722	150		
	Green		756	150		
	Red		698	150		
	Yellow		590	150		
Large	Blue		1289	150		
	Green		1455	150		
	Red		1329	150		
	Yellow		1370	150		
Xlarge	Blue		1498	150		
	Green		1248	150		
	Red		1266	150		
	Yellow		1322	150		
ID	Season	Category	Supplier	Unit Price	Special Price	Discontinued
RMO28497	All	Footwe	7993	\$49.95	\$44.89	No
Description Hiking Walkers with Patterned Tread Durable Uppers						
Size	Color	Style	Units in Stock	Reorder Level	Units on Order	
7	Brown		389	100		
	Tan		422	100		
8	Brown		597	100		
	Tan		521	100		
9	Brown		633	100		
	Tan		654	100		
10	Brown		836	100		
	Tan		954	100		
11	Brown		862	100		
	Tan		792	100		
12	Brown		754	100		
	Tan		788	100		
13	Brown		830	100		
	Tan		921	100		

Drill Down Online Report

Summary and Detailed



Monthly Sales Summary

Year 2013 Month January

Category	Season Code	Web Sales	Telephone Sales	Mail Sales	Total Sales
Footwear	All	\$ 289,323	\$ 1,347,878	\$ 540,883	\$ 2,178,084
Men's Clothing	Spring	\$ 1,768,454	\$ 2,879,243	\$ 437,874	\$ 4,691,484
	Summer	213,938	387,121	123,590	724,649
	Fall	142,823	129,873	112,234	384,930
	Winter	2,980,489	6,453,896	675,290	10,109,675
	All	4,839,729	10,471,135	1,348,987	16,659,851
Totals			\$ 17,473,368	\$ 1,698,222	\$ 23,391,023
Women's Clothing	Spring				965,610
	Summer				
	Fall				
	Winter				
	All				
Totals					

Monthly Sales Detail

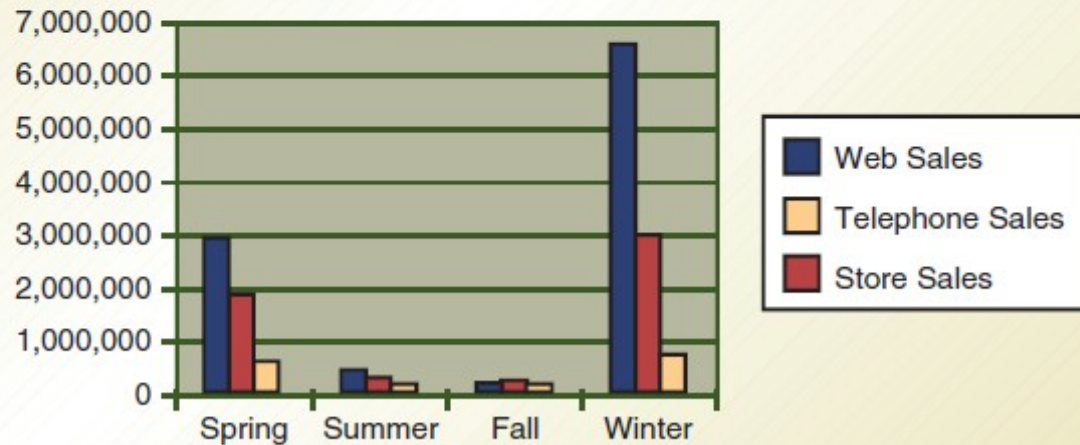
Year	2013	Month	January	Category	Men's Clothing	Season	Winter
Product ID	Product Description		Web Sales	Telephone Sales	Mail Sales	Total Sales	
RMO12987	Winter Parka		\$ 1,490,245	\$ 3,226,948	\$ 337,640	\$ 5,054,833	
RMO13788	Fur-Lined Gloves		149,022	322,695	33,765	505,482	
RMO23788	Wool Sweater		596,097	1,290,775	135,058	2,021,930	
RMO12980	Long Underwear		298,050	645,339	68,556	1,003,005	
RMO32998	Fleece-Lined Jacket		447,075	1,258,079	100,271	1,805,425	
Total			\$ 2,980,489	\$ 6,743,836	\$ 675,290	\$ 10,394,615	

Graphical Outputs

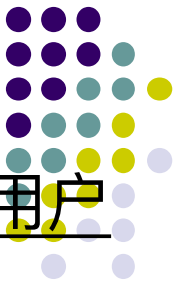
Men's Clothing Sales - January 2013



Men's Clothing Sales by Season - January 2013

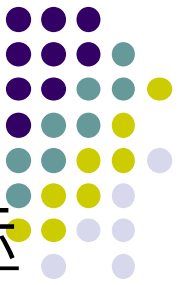


Summary



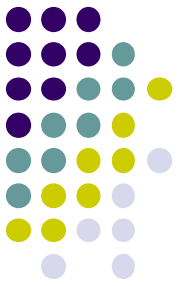
- There are two types of interfaces – user interfaces (用户接口) and system interfaces (系统接口)
- User interfaces involve **direct user interaction with the system** (用户与系统的直接交互). System interfaces require minimal or no user interaction
- The design of the user interface has a long history as human computer interaction (HCI) and relies on user-centered design, which focuses early on users, evaluates designs to ensure usability, and uses iterative development (用户界面的设计作为人机交互 (HCI) 有着悠久的历史，它依赖于以用户为中心的设计)

Summary (continued)



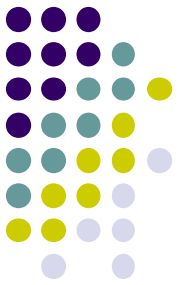
- Key user interface concepts include affordance (提示性) and visibility (可见性) for controls
- Other key principles include consistency, shortcuts, feedback, dialog closure, error handling, reversal of actions, and reducing short term memory loads (其他准则：一致性、快捷方式、反馈、对话框关闭、错误处理、操作反转和减少短期记忆负荷)
- Use cases are organized into one or more menu hierarchies to arrange functionality for users (窗体设计原则：界面布局，数据输入，导航与支持控件)
- Guidelines are available for designing for Windows, Web browsers, and Handheld devices (手持设备：小屏幕，小键盘和触摸屏，有限的网络容量)

Summary (continued)



- System interfaces include inputs and outputs to other systems, highly automated inputs and outputs, and inputs and outputs to external databases. (系统接口包括对其他系统的输入和输出、高度自动化的输入和输出以及对外部数据库的输入和输出。)
- Designing system **inputs** involves identifying devices and mechanisms, identifying inputs and the data content, and determining the controls necessary (设计系统输入包括识别设备和机制，识别输入和数据内容，以及确定必要的控制)
- Designing system **outputs** includes designing detailed reports, summary report, exception reports, and executive reports (设计系统输出包括设计详细报告、摘要报告、异常报告和执行报告)

Summary (continued)



- Outputs are also classified as internal, external, or turnaround (输出也分为内部、外部或周转)
- Electronic reports and other outputs can include drill down, graphics, and multimedia (电子报告和其他输出可以包括深挖报告、图形和多媒体)